Example 8.7

What is the pH of an aqueous solution obtained by mixing 6 gram of acetic acid and 8.2 gram of sodium acetate making the volume equal to 500 ml. (Given: K_a for acetic acid is 1.8×10^{-5})

According to Henderson – Hasselbalch equation,

$$pH = pK_a + \log \frac{[salt]}{[acid]}$$

$P^{K_a} = -\log K_a = -\log(1.8 \times 10^{-5}) = 4.74$ (Refer previous example)

$$[Salt] = \frac{\text{Number of moles of sodium acetate}}{\text{Volume of the solution (litre)}}$$

$$\text{Number of moles of sodium acetate} = \frac{\text{mass of sodium acetate}}{\text{molar mass of sodium acetate}}$$

$$= \frac{8.2}{82} = 0.1$$

$$\therefore [Salt] = \frac{0.1 \text{ mole}}{1/2 \text{ Litre}} = 0.2M$$

$$[acid] = \frac{\left(\frac{\text{mass of } CH_3COOH}{\text{molar mass of } CH_3COOH}\right)}{\text{Volume of solution in litre}}$$

$$= \frac{\left(\frac{6}{60}\right)}{1/2}$$

$$= 0.2M$$

$$\therefore pH = 4.74 + \log \frac{(0.2)}{(0.2)}$$

$$pH = 4.74 + \log 1$$

$$pH = 4.74 + 0 = 4.74$$

Evaluate yourself - 9

a) How can you prepare a buffer solution of pH 9. You are provided with 0.1M NH_4OH solution and ammonium chloride crystals. (Given: pK_b for NH_4OH is 4.7 at 25°C.

b) What volume of 0.6M sodium formate solution is required to prepare a buffer solution of pH 4.0 by mixing it with 100ml of 0.8M formic acid. (Given: pK_a for formic acid is 3.75.)

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